
Software Requirements Specification

for

Multi-Agent Workforce Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for the software system titled “A Multi-Agent System for Employee Performance and Appraisal”, Release 1.0.

The purpose of this document is to provide a detailed description of the functional and non-functional requirements of the proposed system, serving as a reference for developers, project managers, evaluators, and other stakeholders. This SRS outlines what the system is expected to do, without specifying implementation details.

The scope of this product includes an intelligent, data-driven platform that automates employee attendance logging, daily work tracking, timesheet management, performance evaluation, and appraisal calculation using a multi-agent system architecture. The system aims to ensure transparency, fairness, and objectivity in employee performance assessment while minimizing manual intervention and favoritism.

This SRS covers the complete software system, including employee, manager, and administrative functionalities, as well as the interactions among autonomous agents responsible for attendance monitoring, task tracking, contribution analysis, performance evaluation, and appraisal computation. Hardware-level components and external payroll or tax systems, if any, are considered outside the scope of this document.

1.2 Document Conventions

This SRS document follows the guidelines and structure recommended by the IEEE Software Requirements Specification (SRS) standard.

The following conventions are used throughout this document:

- *All requirements are stated using clear, concise, and unambiguous language.*
- *Mandatory requirements are expressed using the keyword “shall”, while optional or recommended features are described using “should” or “may”.*
- *Section and subsection numbering follows the IEEE hierarchical format (e.g., 1.1, 2.3.1).*
- *Functional and non-functional requirements are listed separately and uniquely identifiable.*
- *Role names such as Employee, Manager, and Administrator are capitalized to indicate system-defined user roles.*
- *Performance metrics, constraints, and system limits are explicitly mentioned where applicable.*
- *Priorities for high-level requirements are assumed to be inherited by the detailed requirements under them unless otherwise specified.*

This document uses standard formatting without special typography or color-coding. Any abbreviations or technical terms used are defined in the Definitions, Acronyms, and Abbreviations section of the SRS.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) document is intended for a diverse group of stakeholders involved in the development, evaluation, deployment, and usage of the Multi-Agent System for Employee Performance and Appraisal.

The intended audience includes:

- Developers and System Architects, who will use this document to understand the functional behavior, system features, and constraints required for designing and implementing the software.*
- Project Managers, who will reference this SRS to plan development activities, allocate resources, and track project progress.*
- Testers and Quality Assurance Personnel, who will use the requirements defined in this document to create test cases and validate that the implemented system meets all specified requirements.*
- Evaluators and Academic Reviewers, who will assess the completeness, feasibility, and technical soundness of the proposed system.*
- End Users, including Employees and Managers, who may refer to this document for a high-level understanding of system capabilities and responsibilities.*

This SRS is organized to first provide an overview of the system, followed by a detailed description of system features and requirements. It includes sections on overall system description, functional requirements, external interfaces, non-functional requirements, and future enhancements.

Readers are advised to begin with Section 1 (Introduction) to understand the purpose, scope, and context of the system. Section 2 (Overall Description) provides a conceptual overview of system functionality and operating environment. Section 3 (System Features and Functional Requirements) should be reviewed by developers and testers for detailed behavior. Section 4 and 5, covering external interfaces and non-functional requirements, are particularly relevant for developers and system designers, while future scope sections may be useful for planners and reviewers.

1.4 Product Scope

The Multi-Agent System for Employee Performance and Appraisal is an intelligent software solution designed to objectively monitor employee productivity, attendance, and work contributions in a project-driven organizational environment. The system automates daily attendance tracking, timesheet logging, task monitoring, and managerial approval processes using autonomous software agents.

The primary purpose of the product is to provide a transparent, data-driven, and unbiased employee performance evaluation and appraisal mechanism. By correlating work output, time investment, and contribution data, the system eliminates manual dependency and favoritism in appraisals while ensuring accountability at both employee and managerial levels.

Key objectives and benefits of the system include:

- Improving fairness and accuracy in employee appraisals*
- Enhancing organizational productivity through continuous performance tracking*
- Reducing administrative overhead related to attendance, timesheets, and reviews*
- Supporting informed managerial decisions through real-time analytics*

- *Aligning employee effort directly with appraisal outcomes and compensation*

The system supports organizational goals such as performance optimization, workforce transparency, and operational efficiency, making it suitable for corporate environments, software development teams, and project-based organizations. This product contributes to strategic business goals by enabling measurable performance assessment and consistent workforce management.

If a separate vision or scope document is maintained, this SRS serves as the authoritative source for detailed software requirements and operational boundaries without duplicating higher-level strategic documentation.

1.5 References

The following documents and standards are referenced in the preparation and understanding of this Software Requirements Specification (SRS) for the Multi-Agent System for Employee Performance and Appraisal:

1. *IEEE Std 830-1998
IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers, 1998.*
2. *IEEE Std 29148-2018
ISO/IEC/IEEE International Standard – Systems and Software Engineering — Life Cycle Processes — Requirements Engineering, IEEE Computer Society, 2018.*
3. *Wooldridge, M.
An Introduction to MultiAgent Systems, Second Edition, John Wiley & Sons, 2009.*
4. *Pressman, R. S., & Maxim, B. R.
Software Engineering: A Practitioner's Approach, 8th Edition, McGraw-Hill Education, 2014.*
5. *Git Documentation
Git – Distributed Version Control System, Software Freedom Conservancy, available online.*
6. *ISO/IEC 25010:2011
Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE), International Organization for Standardization, 2011.*

These references provide the standards, theoretical foundations, and technical guidelines used to define the requirements and structure of the proposed system.

2. Overall Description

2.1 Product Perspective

The Multi-Agent System for Employee Performance and Appraisal is a new, self-contained software product designed to address limitations in traditional employee performance evaluation

and appraisal systems. It is not a direct replacement for any single existing software but rather an integrated solution that automates and improves multiple manual and semi-automated processes such as attendance tracking, timesheet management, performance evaluation, and appraisal calculation.

The product is designed to operate as an independent system while supporting integration with external tools such as version control systems (e.g., Git repositories) for analyzing employee work contributions. The system adopts a multi-agent architecture, where each autonomous agent is responsible for a specific operational domain, such as attendance monitoring, work tracking, managerial approval, or performance evaluation.

At a high level, the system consists of the following major components:

- User Interface for Employees, Managers, and Administrators
- Attendance and Timesheet Management Module
- Task and Work Item Management Module
- Git Contribution Tracking Module
- Performance Evaluation and Appraisal Module
- Salary Computation Module
- Multi-Agent Coordination Layer
- Centralized Data Storage and Reporting Module

The agents collaborate by sharing relevant data through a common data repository and communication interfaces. External systems, such as Git hosting services, serve as data sources for contribution analysis but are not controlled by this system.

This SRS defines the requirements for the complete software system, including internal agent interactions and user-facing functionalities. Any external enterprise systems such as payroll tax systems or biometric hardware are considered outside the scope of this specification.

2.2 Product Functions

At a high level, the Multi-Agent System for Employee Performance and Appraisal provides the following major functions:

- User authentication and role-based access control for Employees, Managers, and Administrators
- Automated employee attendance tracking, including check-in and check-out
- Daily timesheet logging and work description entry
- Task and work item management with status tracking
- Integration with Git repositories to monitor code contributions
- Managerial review, approval, and feedback on employee work
- Objective productivity and performance evaluation using defined metrics
- Automated appraisal score calculation based on approved work and attendance
- Monthly salary computation based on performance and attendance data
- Generation of reports and dashboards for performance, attendance, and appraisal
- Notification and alert generation for approvals, deadlines, and anomalies
- Secure storage and audit logging of all system activities

Detailed descriptions of these functions, along with their inputs, outputs, and constraints, are provided in

2.3 User Classes and Characteristics

The Multi-Agent System for Employee Performance and Appraisal is intended to be used by multiple classes of users, each with different responsibilities, access privileges, and interaction levels. The system enforces role-based access control to ensure security and appropriate functionality for each user class.

Employee

Employees are the primary users of the system. They interact with the software on a daily basis to record attendance, log work details, and monitor their own performance metrics. Employees are expected to have basic computer literacy and familiarity with task-based work environments. Their access is limited to personal data, assigned tasks, submitted timesheets, and individual performance reports. Employees do not have permission to modify appraisal rules or access other users' data.

Manager

Managers are frequent users responsible for overseeing employee work and validating submitted timesheets and tasks. They review daily or weekly work logs, provide feedback, and approve or reject work entries. Managers have access to performance summaries and analytics for their respective teams but cannot alter system-wide configurations. Managers are assumed to have moderate technical proficiency and familiarity with project management practices.

Administrator

Administrators are less frequent but critical users of the system. They are responsible for configuring system parameters, managing user accounts, defining appraisal metrics, and maintaining access control policies. Administrators have full system privileges, including access to all reports, logs, and configuration settings. They are expected to have higher technical expertise and system management experience.

System Agents

System Agents are autonomous, non-human entities that perform background operations such as attendance tracking, timesheet validation, contribution analysis, performance evaluation, and salary computation. These agents operate continuously without direct user interaction and are critical to system functionality. While not end users, they are essential actors within the system.

Among these user classes, Employees and Managers are the most important for usability and functional satisfaction, as they directly interact with the system on a daily basis. Administrator usability is important but secondary, while System Agents are focused on correctness, reliability, and performance rather than user experience.

2.4 Operating Environment

The Multi-Agent System for Employee Performance and Appraisal is designed to operate in a standard web-based client-server environment.

The system will run on the following platforms and environments:

- **Client Side:**
Any modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge, running on desktop or laptop systems.

- **Server Side:**
A server environment supporting web application deployment, running on a standard operating system such as Windows or Linux.
- **Operating Systems:**
Windows 10 or higher, Linux-based operating systems, or equivalent server operating systems.
- **Backend and Supporting Software:**
 - *Web server (e.g., Apache or equivalent)*
 - *Application server supporting the chosen programming language*
 - *Database management system for persistent data storage*
 - *Git hosting platforms for repository integration*

The system is required to coexist peacefully with other organizational software such as project management tools, version control systems, and reporting tools. No specialized hardware is required beyond standard computing devices with internet connectivity.

2.5 Design and Implementation Constraints

The design and implementation of the Multi-Agent System for Employee Performance and Appraisal are subject to the following constraints:

- **Architectural Constraints:**
The system shall be designed using a multi-agent architecture, where autonomous agents are responsible for distinct functional areas such as attendance tracking, timesheet validation, contribution analysis, and appraisal computation. This architectural choice limits the system design to agent-based interaction models.
- **Technology Constraints:**
The system shall be implemented using commonly available, open-source technologies suitable for web-based applications. The backend must support agent coordination and data persistence, while the frontend must support role-based user interactions.
- **Integration Constraints:**
The system must interface with external version control systems (e.g., Git repositories) to retrieve contribution data. These integrations are subject to the availability and access permissions provided by the external platforms.
- **Security Constraints:**
The system must enforce role-based access control, secure authentication, and data protection mechanisms to prevent unauthorized access to sensitive employee and appraisal data.
- **Performance Constraints:**
The system should efficiently handle concurrent users and background agent activities without significant degradation in response time. Attendance logging and timesheet submissions must be recorded in near real-time.
- **Data Storage Constraints:**
All attendance records, timesheets, approvals, and appraisal data must be persistently stored in a centralized database. Historical data shall not be deleted to ensure auditability.

- **Regulatory and Policy Constraints:**
The system must comply with general organizational policies regarding employee data confidentiality and usage. Domain-specific legal or taxation requirements related to payroll are outside the scope of this system.
- **Maintenance Constraints:**
The system shall follow standard coding and documentation practices to ensure maintainability, as future maintenance may be handled by a different development team.

2.6 User Documentation

The following user documentation components will be delivered along with the Multi-Agent System for Employee Performance and Appraisal:

- **User Manual:**
A comprehensive user manual describing system features, navigation, and usage instructions for Employees, Managers, and Administrators.
- **Online Help Documentation:**
Context-sensitive help integrated within the system to assist users during common operations such as attendance logging, timesheet submission, and approval.
- **Administrator Guide:**
Documentation detailing system configuration, user management, appraisal rule definition, and maintenance procedures for Administrators.
- **Quick Start Guide:**
A concise guide providing basic steps for first-time users to start using the system.

The documentation will be provided in electronic format (PDF and/or HTML) and written in clear, user-friendly language. All documentation will conform to standard software documentation guidelines to ensure clarity and consistency.

2.7 Assumptions and Dependencies

The requirements specified in this Software Requirements Specification (SRS) are based on the following assumptions and dependencies. Changes to these assumptions may impact system functionality, design decisions, or implementation feasibility.

Assumptions

- *It is assumed that all users (Employees, Managers, and Administrators) have access to a stable internet connection and a compatible web browser.*
- *Employees are assumed to accurately and honestly log their daily work activities and timesheets.*
- *Managers are assumed to regularly review and approve or reject submitted work within the defined timeframes.*
- *It is assumed that organizational policies governing attendance, performance evaluation, and appraisal metrics are clearly defined and consistent.*
- *The system assumes availability of valid access credentials for integrating with external Git repositories when contribution tracking is enabled.*

- It is assumed that system usage will comply with organizational data privacy and confidentiality policies.

Dependencies

- The system depends on external version control platforms (such as Git-based repositories) to retrieve contribution data. Any changes in external APIs, authentication mechanisms, or access permissions may affect functionality.
- The performance evaluation and appraisal computation depend on the accuracy and completeness of attendance records, timesheets, and managerial approvals.
- The system depends on a centralized database for storing and retrieving all operational data; database availability and reliability are critical to system operation.
- The multi-agent system depends on the underlying application framework to support concurrent background processes and inter-agent communication.
- Deployment and operation of the system depend on the availability of the required server infrastructure and supporting software components.

These assumptions and dependencies will be reviewed and validated during system design and implementation to minimize risk and ensure alignment with real-world operating conditions.

3. External Interface Requirements

3.1 User Interfaces

The Multi-Agent System for Employee Performance and Appraisal provides a role-based, web-based graphical user interface (GUI) to support interaction between the software and its users. The user interface is designed to be intuitive, consistent, and accessible across different user roles.

General User Interface Characteristics

- The system shall provide a web-based interface accessible through standard web browsers.
- All users shall authenticate through a secure login interface.
- The interface shall adapt dynamically based on the user role (Employee, Manager, or Administrator).
- Common interface elements such as navigation menus, action buttons (Submit, Approve, Reject), and help options shall be consistent across all screens.
- Standard system messages, confirmations, and error notifications shall be displayed clearly and unambiguously.

Employee Interfaces

- Attendance interface for daily check-in and check-out.
- Timesheet interface for logging daily work and task details.
- Task and work item view displaying assigned work and status.
- Performance and appraisal dashboard showing productivity metrics and evaluation results.
- Notification panel for alerts and manager feedback.

Manager Interfaces

- Dashboard displaying team attendance and work summaries.
- Timesheet and task approval interface with feedback options.

- Performance analytics and appraisal summary views for managed employees.
- Notification interface for pending approvals and system alerts.

Administrator Interfaces

- User and role management interface.
- Configuration screens for appraisal metrics and system parameters.
- System monitoring and reporting interfaces.
- Audit log and activity tracking views.

Detailed screen layouts, workflows, and visual design guidelines will be documented in a separate User Interface Specification and are outside the scope of this SRS.

3.2 Hardware Interfaces

The Multi-Agent System for Employee Performance and Appraisal does not require direct interaction with specialized hardware components. The system is designed to operate using standard computing hardware.

- The software shall operate on standard desktop and laptop computers.
- User access requires basic input devices such as a keyboard, mouse, or trackpad.
- The system shall support execution on servers with standard processing, memory, and storage capabilities sufficient for web-based applications.
- No biometric devices, sensors, or dedicated hardware interfaces are required for system operation.

If deployed in environments where additional hardware (such as biometric attendance devices) is used, such integrations are considered outside the scope of this SRS and may be addressed in future enhancements.

3.3 Software Interfaces

The Multi-Agent System for Employee Performance and Appraisal interfaces with several software components to provide its intended functionality. These interfaces enable data storage, processing, and integration with external systems.

Database Interface

- The system shall interface with a centralized database management system for persistent storage of:
 - User profiles and role information
 - Attendance logs and timesheets
 - Task and work item records
 - Manager approvals and feedback
 - Performance metrics and appraisal results
 - Audit and activity logs
- The database interface shall support secure read and write operations initiated by both user actions and autonomous agents.

Operating System Interface

- The system shall operate on standard server operating systems such as Windows or Linux.
- Background agents shall rely on operating system services for task scheduling, process management, and resource allocation.

Version Control System Interface

- The system shall interface with external Git-based version control platforms to retrieve contribution data such as commits, pull requests, and timestamps.
- Data received from the version control system is used by the contribution analysis agents for performance evaluation.
- The interface shall be read-only and dependent on access credentials and permissions provided externally.

Application Framework and Libraries

- The system shall utilize standard web application frameworks and supporting libraries for user interface rendering, authentication, and agent communication.
- Inter-agent communication shall occur through shared data storage or internal messaging services.

Data Exchange

- Data exchanged between internal components includes attendance records, timesheet entries, approval status, contribution summaries, and evaluation scores.
- Data shared across software components shall maintain consistency and integrity through controlled access mechanisms.
- Detailed API specifications and integration protocols for external software components will be documented separately and are not part of this SRS.

3.4 Communications Interfaces

The Multi-Agent System for Employee Performance and Appraisal requires network-based communication to support user access and integration with external services.

- The system shall support communication between client browsers and the server using standard HTTP/HTTPS protocols.
- Secure communication shall be enforced using encryption mechanisms to protect sensitive data during transmission.
- The system shall support asynchronous communication for background agent operations without affecting user interaction responsiveness.
- Notification services may use internal messaging or email protocols for alert delivery.

Communication Standards and Security

- All user authentication and data exchange shall occur over secure channels.
- The system shall ensure data integrity and confidentiality during communication.
- Session management and timeout mechanisms shall be implemented to prevent unauthorized access.

Data Synchronization

- Communication between agents and system components shall ensure synchronized and consistent data updates.

- Conflict resolution mechanisms shall be applied where concurrent updates may occur.

These communication interfaces ensure reliable, secure, and efficient data exchange among users, agents, and external systems.

4. System Features

This section describes the major system features of the Multi-Agent System for Employee Performance and Appraisal. Each feature represents a high-level service provided by the system. Detailed functional requirements are stated using “shall” statements to define mandatory system behavior. The features are organized based on functional hierarchy and user interaction.

4.1 Attendance Management

4.1.1 Description and Priority

The Attendance Management feature enables the system to automatically record and manage employee attendance through daily check-in and check-out actions. This feature captures working hours, supports attendance-based productivity tracking, and provides reliable input for performance evaluation and appraisal calculations. Attendance data is processed by autonomous agents and stored securely for reporting and audit purposes.

Priority: High

Benefit: 9 (Critical for appraisal accuracy)

Penalty if absent: 9 (System loses credibility and fairness)

Cost: 4 (Moderate implementation complexity)

Risk: 3 (Low technical risk)

Stimulus/Response Sequences

4.1.2 Stimulus/Response Sequences

Stimulus 1: Employee logs into the system at the start of the workday.

Response: The system displays the attendance interface with a check-in option.

Stimulus 2: Employee performs a check-in action.

Response: The system records the check-in time, confirms successful attendance logging, and updates the attendance database.

Stimulus 3: Employee performs a check-out action at the end of the workday.

Response: The system records the check-out time, calculates total working hours, and stores the attendance record.

Stimulus 4: Manager accesses attendance records.

Response: The system displays attendance summaries for assigned employees.

Stimulus 5: Employee attempts an invalid operation (e.g., multiple check-ins without check-out).

Response: The system displays an appropriate error message and prevents inconsistent data entry.

4.1.3 Functional Requirements

REQ-ATT-1:

The system shall allow employees to check in once per working day.

REQ-ATT-2:

The system shall allow employees to check out once per working day after a valid check-in.

REQ-ATT-3:

The system shall record the exact timestamp for every check-in and check-out action.

REQ-ATT-4:

The system shall automatically calculate total working hours based on recorded attendance timestamps.

REQ-ATT-5:

The system shall prevent duplicate or invalid check-in and check-out attempts.

REQ-ATT-6:

The system shall store all attendance records in a persistent database.

REQ-ATT-7:

The system shall make attendance data available to performance evaluation and appraisal agents.

REQ-ATT-8:

The system shall allow managers to view attendance summaries of their assigned employees.

REQ-ATT-9:

The system shall generate error messages for missing, invalid, or incomplete attendance actions.

4.2 Timesheet and Work Logging.

4.2.1 Description and Priority

This feature allows employees to record daily work activities through structured timesheet entries. Timesheets capture task details, time spent, and project association. The collected data serves as a primary input for productivity measurement and performance appraisal.

Priority: High

Benefit: 9

Penalty if absent: 8

Cost: 5

Risk: 3

4.2.2 Stimulus/Response Sequences

Stimulus 1: Employee selects “Add Timesheet”.

Response: System displays a timesheet entry form.

Stimulus 2: Employee submits timesheet details.
Response: System validates data and stores the timesheet.

Stimulus 3: Employee edits a pending timesheet.
Response: System updates the record before approval.

Stimulus 4: Invalid or missing input.
Response: System displays validation errors.

4.2.3 Functional Requirements

REQ-TS-1: The system shall allow employees to submit daily timesheets.
REQ-TS-2: The system shall associate timesheets with projects and tasks.
REQ-TS-3: The system shall validate time duration and mandatory fields.
REQ-TS-4: The system shall allow editing of timesheets before approval.
REQ-TS-5: The system shall prevent modification after approval.
REQ-TS-6: The system shall store timesheet history permanently.

4.3 Task and Work Item Management

4.3.1 Description and Priority

This feature manages creation, assignment, and tracking of tasks and work items within projects.
Priority: High
Benefit: 8
Penalty if absent: 7
Cost: 5
Risk: 4

4.3.2 Stimulus/Response Sequences

Stimulus 1: Manager creates a task.
Response: Task is assigned to employee.

Stimulus 2: Employee updates task status.
Response: System records task progress.

4.3.3 Functional Requirements

REQ-TASK-1: The system shall allow managers to create tasks.
REQ-TASK-2: The system shall assign tasks to employees.
REQ-TASK-3: The system shall track task status.
REQ-TASK-4: The system shall associate tasks with timesheets.
REQ-TASK-5: The system shall record task completion dates.

4.4 Manager Review and Approval

4.4.1 Description and Priority

This feature enables managers to validate employee work through approval or rejection of timesheets and tasks.

Priority: High

Benefit: 9

Penalty if absent: 9

Cost: 4

Risk: 3

4.4.2 Stimulus/Response Sequences

Stimulus 1: Manager views pending approvals.

Response: System displays submitted work.

Stimulus 2: Manager approves or rejects work.

Response: System updates status and notifies employee.

4.4.3 Functional Requirements

REQ-APR-1: The system shall allow managers to review timesheets.

REQ-APR-2: The system shall allow approval or rejection.

REQ-APR-3: The system shall capture manager feedback.

REQ-APR-4: The system shall notify employees of decisions.

REQ-APR-5: The system shall log all approval actions.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall meet the following performance requirements to ensure efficient and reliable operation:

PERF-1: The system shall respond to user interface actions (such as login, attendance marking, and timesheet submission) within **2 seconds** under normal operating conditions.

PERF-2: The system shall support concurrent access by multiple users without data inconsistency or significant performance degradation.

PERF-3: The system shall record attendance check-in and check-out events in **near real-time** with a delay not exceeding **1 second**.

PERF-4: The system shall process performance evaluation and appraisal calculations within **10 seconds** for a standard employee dataset.

PERF-5: The system shall ensure background agents operate continuously without blocking or delaying user-initiated operations.

PERF-6: The system shall maintain acceptable performance levels with increasing numbers of users and stored records through scalable design.

Rationale:

These requirements ensure that the system remains responsive, reliable, and capable of handling real-world organizational workloads efficiently.

5.2 Safety Requirements

Although the system does not control physical hardware, it must ensure safety related to data integrity and system operation.

SAFE-1: The system shall prevent loss of critical data such as attendance records, timesheets, and appraisal results.

SAFE-2: The system shall ensure that unauthorized users cannot modify or delete performance or salary records.

SAFE-3: The system shall maintain audit logs of all critical actions, including attendance recording, approvals, and appraisal computations.

SAFE-4: The system shall prevent invalid operations that could corrupt system data.

SAFE-5: The system shall provide error handling mechanisms to prevent system crashes and ensure graceful recovery.

SAFE-6: The system shall protect appraisal and salary data from accidental or intentional misuse.

5.3 Security Requirements

The system shall ensure secure access, authentication, and protection of sensitive employee data.

SEC-1: The system shall require user authentication through valid login credentials.

SEC-2: The system shall enforce role-based access control to restrict unauthorized access to system functions.

SEC-3: The system shall ensure that employees can access only their own performance and appraisal data.

SEC-4: The system shall ensure that managers can access only data related to their assigned employees.

SEC-5: The system shall protect all sensitive data stored in the database from unauthorized access.

SEC-6: The system shall ensure secure communication between client and server using secure communication protocols.

SEC-7: The system shall maintain logs of login attempts and system access for audit and security monitoring.

SEC-8: The system shall prevent unauthorized modification of appraisal and salary computation results.

5.4 Software Quality Attributes

The system shall meet the following quality standards:

5.4.1 Reliability

QUAL-REL-1: The system shall operate reliably without unexpected failures during normal usage.

QUAL-REL-2: The system shall maintain consistent and accurate data across all modules and agents.

5.4.2 Usability

QUAL-USE-1: The system shall provide an intuitive and user-friendly interface.

QUAL-USE-2: Users shall be able to perform common operations without requiring specialized technical knowledge.

5.4.3 Maintainability

QUAL-MAIN-1: The system shall be designed using modular architecture to allow easy maintenance and updates.

QUAL-MAIN-2: The system shall provide clear documentation for developers and administrators.

5.4.4 Scalability

QUAL-SCAL-1: The system shall support increasing numbers of users, tasks, and projects without requiring major redesign.

5.4.5 Availability

QUAL-AVL-1: The system shall be available for use at least **99% of the operational time**, excluding scheduled maintenance.

5.4.6 Integrity

QUAL-INT-1: The system shall ensure accuracy and consistency of stored data.

5.4.7 Flexibility

QUAL-FLX-1: The system shall allow modification of appraisal rules and evaluation criteria without major system redesign.

5.5 Business Rules

The system shall enforce the following business rules:

BR-1: Employees must check in before submitting timesheets for the day.

BR-2: Employees shall not modify timesheets after manager approval.

BR-3: Managers must approve or reject employee timesheets before appraisal calculations.

BR-4: Appraisal scores shall be calculated based only on approved attendance, tasks, and contributions.

BR-5: Only authorized administrators shall modify system configuration settings.

BR-6: Salary computation shall be based on verified attendance, approved work, and performance evaluation.

BR-7: The system shall maintain a complete audit trail of all appraisal and approval activities.

BR-8: Autonomous agents shall operate according to predefined system rules and shall not override manual approvals.